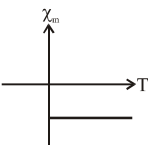
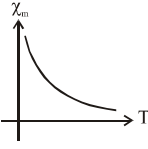
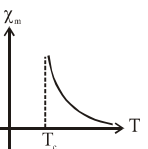


MAGNETISM & MATTER

1. Unit of magnetic flux density is :-
 (1) Tesla (2) Wb/m^2
 (3) $\text{NA}^{-1} \text{m}^{-1}$ (4) All of the above
2. If a magnet of pole strength m is cut into four parts such that length and width of each part is half that of initial one, then pole strength of each part :-
 (1) $m/4$ (2) m (3) $m/8$ (4) $m/2$
3. Magnetic field at a distance d from a short bar magnet in $\tan A$ and $\tan B$ position are in ratio:-
 (1) 2 : 1 (exact)
 (2) 2 : 1 (approx)
 (3) 1 : 2 (exact)
 (4) 1 : 2 (approx)
4. A magnet of magnetic moment M is situated having its axis along magnetic field B , the work done in rotating the magnet by 180° will be:-
 (1) $-MB$ (2) $+MB$ (3) zero (4) $+2MB$
5. Material suitable for making electromagnet should have :-
 (1) High retentivity and High coercivity
 (2) Low retentivity and Low coercivity
 (3) High retentivity and Low coercivity
 (4) Low retentivity and High coercivity
6. A dip circle is at right angle to magnetic meridian, then apparent dip :-
 (1) 0° (2) 30° (3) 60° (4) 90°
7. A line passing through places having zero dip angle is called :-
 (1) Isoclinic lines (2) Aclinic lines
 (3) Isogonic lines (4) Agonic lines
8. Universal property of all substance is :-
 (1) Diamagnetism
 (2) Paramagnetism
 (3) Ferromagnetism
 (4) All of above
9. Unit of magnetic susceptibility is :-
 (1) Henry (2) Wb/m
 (3) Amp/m (4) None of these
10. Magnetic field is measured by :-
 (1) Pyrometer (2) Hydrometer
 (3) Thermometer (4) Fluxmeter
11. The magnetic moment produced in a substance of 1 gm is $6 \times 10^{-7} \text{ A}\cdot\text{m}^2$. If its density is 5 gm/cm^3 , then the intensity of magnetisation in A/m will be:-
 (1) 8.3×10^6 (2) 3.0
 (3) 1.2×10^{-7} (4) 3×10^{-6}
12. Relative permeability of iron is 5500, then its magnetic susceptibility is :-
 (1) 5500 (2) 5501
 (3) 5499 (4) zero
13. Superconducting material is :-
 (1) Diamagnetic (2) Perfect diamagnetic
 (3) Paramagnetic (4) Ferromagnetic
14. If current is doubled, the deflection is also doubled in :-
 (1) Tangent galvanometer
 (2) Moving coil galvanometer
 (3) both (1) and (2)
 (4) None of above
15. For paramagnetic materials :-
 (1) χ is positive at all temperature
 (2) χ is negative at all temperature
 (3) χ may be positive or negative
 (4) χ does not depend on temperature
16. Magnetic field lines represent the direction :-
 (1) Along which a small magnetised needle aligns
 (2) Along which moving charge particle experiences a force
 (3) Both (1) and (2)
 (4) None of (1) and (2)
17. Which effect is responsible for earth magnetic field :-
 (1) Dynamo effect (2) Photo electric effect
 (3) Compton effect (4) Solar effect
18. When a dip magnetic needle is suspended in the earth's magnetic field :-
 (1) In northern hemisphere, the north pole of dip tilts downwards. At southern hemisphere south pole of dip tilts downwards
 (2) In northern hemisphere, the south pole of dip tilts downwards. At southern hemisphere north pole of dip tilts downwards
 (3) In both (northern and southern hemisphere), north pole of dip tilts downwards.
 (4) In both northern, hemisphere and southern hemisphere, south-pole of dip tilts downwards

19. The positions on earth where angle of declination is greater, at :-
 (1) Higher latitudes
 (2) Near the equator
 (3) Lower latitudes
 (4) Same at all position of the earth
20. The position on the earth where angle of declination is smaller, at :-
 (1) Higher latitudes
 (2) Near the equator
 (3) Away from equator
 (4) Same at all positions of earth
21. In Delhi and Mumbai magnetic needle shows the true north quite accurately because :-
 (1) In Delhi and Mumbai angle of dip is small
 (2) In Delhi and Mumbai angle of dip is large
 (3) In Delhi and Mumbai angle of declination is small
 (4) In Delhi and Mumbai angle of declination is large
22. When a superconductor is placed near a bar magnet then :-
 (1) It repels the magnet
 (2) It attracts the magnet
 (3) Neither repels nor attracted
 (4) Some time repels and some time attracts
23. Dia-magnetic property of material can be explained by :-
 (1) Lenz's law (2) Faraday law
 (3) Ampere's law (4) Gauss's law
24. A ferromagnetic material is placed in an external magnetic field. The magnetic domains:
 (1) increase in size
 (2) decrease in size
 (3) may increase or decrease in size
 (4) have no relation with the field
25. If magnetic monopoles existed, how would the Gauss's law of magnetism be modified ?
 (Here q_m is the monopole magnetic charge enclosed by surface S)
 (1) $\oint \vec{B} \cdot d\vec{S} = \frac{q_m}{\mu_0}$ (2) $\oint \vec{B} \cdot d\vec{S} = q_m$
 (3) $\oint \vec{B} \cdot d\vec{S} = \mu_0 q_m$ (4) $\oint \vec{B} \cdot d\vec{S} = \mu_0 q_m^2$
26. Which of the following statements are correct?
 (A) Magnetic field lines can be entirely confined (within the core) of a toroid
 (B) Magnetic field lines can be entirely confined within the core of a solenoid
 (C) a bar magnet exert a torque on itself due to its own field.
 (D) a system can have magnetic moment even though its net charge is zero.
 (1) A, B, D (2) only A, D
 (3) A, C (4) A, B, C, D
27. Which planets have maximum & minimum magnetic fields respectively :-
 (1) Jupiter & Venus
 (2) Mercury & Mars
 (3) Jupiter & Mars
 (4) Venus & Mercury
28. A solenoid has core of material with relative permeability 400. The winding of the solenoid are insulated from the core and carry a current of 2A. It has 1000 turns/meter. Then magnetising current will be :-
 (1) 798 A (2) 1000 A
 (3) 500 A (4) 1494 A
29. The earth's magnetic field at the equator is approximately 0.4 G. Then the earth's dipole moment will be :-
 (1) $1.05 \times 10^{23} \text{ A-m}^2$ (2) $5.05 \times 10^{23} \text{ A-m}^2$
 (3) $9.5 \times 10^{23} \text{ A-m}^2$ (4) $1.05 \times 10^{21} \text{ A-m}^2$
30. If external magnetic intensity on a paramagnetic material is made 4 times and the absolute temperature is made 3 times then how many times will be the self magnetisation ?
 (1) 4 times (2) $\frac{1}{3}$ times
 (3) $\frac{4}{3}$ times (4) $\frac{3}{4}$ times
31. The angle of declination is :-
 (1) The angle of earth magnetic field with horizontal
 (2) The angle of earth magnetic field with vertical
 (3) The angle between the geographic axis and magnetic axis of the earth
 (4) The angle between the geographic meridian and the magnetic meridian

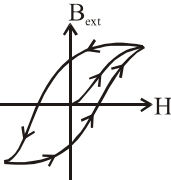
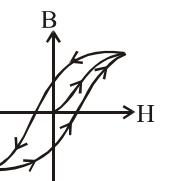
32. Match the column I, in which magnetic susceptibility (χ_m) and temperature (T) curve is given, to the suitable magnetic material of the column-II choose the correct option from the codes given below :-

Column-I	Column-II
(A) 	(p) Ferromagnetic material
(B) 	(q) diamagnetic material
(C) 	(r) Paramagnetic material

Code

A	B	C
(1) p	q	r
(2) r	q	p
(3) q	p	r
(4) q	r	p

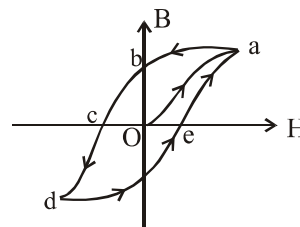
33. For the given uses select the correct magnetic material :-

Column-I	Column-II
(p) Electro magnet	(A) 
(q) Transformer	(B) 
(r) Permanent magnet (B) used in generator	

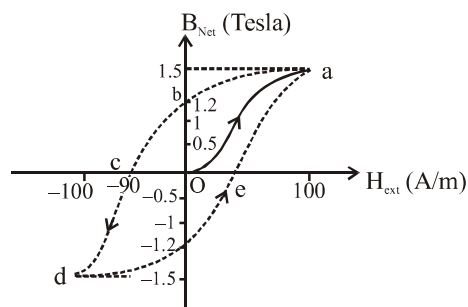
Codes :

p	q	r
(1) A	B	B
(2) B	B	A
(3) B	A	B
(4) A	A	D

34. Figure shows the magnetic hysteresis loop that is the B-H curve for ferromagnetic materials. Select the correct statement :-



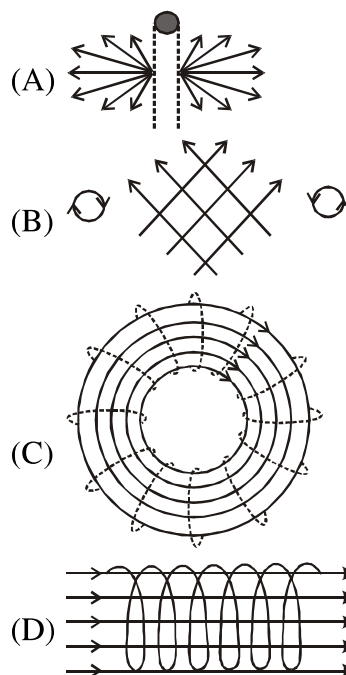
- (1) The value of B at $H = 0$ is called coercivity
 (2) An electromagnet has low remanence and low coercivity
 (3) The value of H at C is called remanence.
 (4) A permanent magnet has low remanence and low coercivity
35. When will the field lines be completely expelled (χ = magnetic susceptibility)
- (1) $\chi = -1$ and $\mu_r = 0$ (2) $-1 \leq \chi$ and $\mu_r \gg 1$
 (3) $\chi \leq 1$ and $\mu_r \ll 1$ (4) $\chi = 1$ and $\mu_r = 1$
36. Select the incorrect statement :-
- (1) When a bar magnet of dipole moment $\vec{M}$ is placed in a uniform magnetic field $\vec{B}$ the torque on it is $\vec{M} \times \vec{B}$
 (2) Gauss's law for magnetism states that the net magnetic flux through any closed surface is zero
 (3) For diamagnetic materials magnetic susceptibility $\chi = +10^{-5}$ whereas paramagnetic materials $\chi = -10^{-5}$
 (4) Magnetisation of a paramagnetic material is inversely proportional to the absolute temperature T.
37. The hysteresis curve for a material is shown in figure. Then for the material retentivity coercivity and saturation magnetic field will be respectively:-



- (1) 90 A/m, 1.2 T, 1.5 T
 (2) 1.5T, 90 A/m, 1.2 T
 (3) 1.2T, 90 A/m, 1.5 T
 (4) 90 A/m, 1.5 T, 1.2 T

38. The magnetic material having very large positive magnetic susceptibility is :-
 (1) Ferromagnetic
 (2) Diamagnetic
 (3) Paramagnetic
 (4) Both Ferromagnetic and Paramagnetic have large positive magnetic susceptibility
39. A long solenoid has 1000 turns per meter and carries a current of 1A. It has a soft iron core of $\mu_r = 1000$. The core is heated beyond the critical temperature, T_c .
 (1) The H field in the solenoid is decreases drastically but the 'B' field is (nearly) unchanged
 (2) The H and B fields in the solenoid are nearly unchanged.
 (3) The magnetisation in the core reverse direction
 (4) The magnetisation in the core diminishes by a factor of about 10^8
40. The line on the earth's surface joining the points where the field is horizontal is called :-
 (1) magnetic meridian
 (2) magnetic axis
 (3) magnetic line
 (4) magnetic equator
41. The magnetic field is now thought to arise due to electrical currents produced by convective motion of metallic fluid. (consisting mostly of molten iron and nickel) in the outer core of the earth. This is known as the
 (1) dynamo effect
 (2) tidal effect
 (3) both (1) and (2)
 (4) None of these

42. Nickel shows ferromagnetic property at room temperature. If the temperature is increased beyond Curie temperature, then it will show :-
 (1) paramagnetism
 (2) ferromagnetism
 (3) no magnetic property
 (4) diamagnetism
43. Many of the diagrams given in figure, show magnetic field lines (thick lines in the figure). Point out which one is/are correct :-



- (1) both A and D (2) both B and D
 (3) both C and D (4) only C
44. New Zealand is situated in southern hemisphere and Russia is situated in Northern hemisphere. The angle of dip at New Zealand and Russia are δ_1 and δ_2 respectively, then :-
 (1) $\delta_1 = +(\text{up})$, $\delta_2 = +(\text{up})$
 (2) $\delta_1 = -(\text{down})$, $\delta_2 = +(\text{up})$
 (3) $\delta_1 = -(\text{up})$, $\delta_2 = +(\text{down})$
 (4) $\delta_1 = -(\text{down})$, $\delta_2 = -(\text{down})$